

Covid-19 pandemic impact on institutional trust

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1. EXECUTIVE SUMMARY

This research investigates the relationship between COVID-19 pandemic and trust within European institutions, considering variables such as trust in the legal system, scientists, and political parties. Our primary hypothesis posits that countries most significantly affected by the pandemic, characterized by high death and case rates, will likely exhibit diminished trust in their institutions. To conduct a thorough analysis of the pandemic's impact on institutional trust in Europe, we selected the 'europecovid.dta' dataset and leveraged data from the European Social Survey (ESS) Round 10, offering valuable insights into the prevailing situation.

A comprehensive cluster analysis was executed to categorize countries based on the extent of their COVID-19 impact and response, considering factors such as death rates, testing levels, stringency indices, and confirmed cases. Subsequently, a factor analysis was applied, extracting three key factors that encapsulate the level of trust in institutions across various European nations.

The outcomes of our study indicate a consistent trend: countries most severely affected by the COVID-19 pandemic generally exhibit lower levels of trust in their institutions. This underscores a critical connection between the pandemic's impact and the erosion of trust.

2. INTRODUCTION

The shock waves of the Covid-19 Pandemic have fundamentally reshaped the landscape of trust in institutions, prompting a critical examination of the dynamic interplay between the crisis and the evolving levels of trust within our societal and political frameworks. In this applied research project, we aim to explore the varying levels of trust within European institutions amid the Covid-19 Pandemic. Our focus is determining if the intensity of the Covid-19 Pandemic is associated with distinct levels of trust in these institutions. The significance of our research question lies in its far-reaching implications for societal resilience and governance. As the Pandemic unfolded, institutions played a central role in managing crises, disseminating information, and implementing public health measures. Understanding how trust in these institutions fluctuated, provides crucial insights into the effectiveness of governance, communication strategies, and crisis management. Moreover, Anthony Giddens theory provides us with a framework to help understand the complexity of institutional trust among different countries. Our research methodology entails a thorough exploration through a comprehensive multivariate analysis utilizing data sourced from the 'eurocovid.dta' and European Social Survey (ESS) Rounds 10. The units of analysis considered, encompass citizens from diverse countries. Rooted in established literature, our hypothesis suggests that political trust is anticipated to correlate with the intensity of the Covid-19 impact on each respective country. Our research strategy involves utilizing a cluster analysis, factor analysis and regression to scrutinize the relationships among variables and substantiate our findings.

3. BACKGROUND

The COVID-19 pandemic, while presenting unprecedented challenges to public health, has sparked profound societal shifts, notably in the realm of political trust. The social landscape, characterized by diverse economic structures and geographic disparities, underwent significant transformations as the pandemic unfolded. Its impact rippled through societies with varying economic vulnerabilities, healthcare infrastructures, and levels of social resilience. Understanding the nuanced interplay among these factors is crucial for deciphering the evolving patterns of political trust. Countries endowed with robust healthcare systems and economic resilience may exhibit distinct trajectories in political trust, standing in contrast to those contending with resource constraints and socio-economic disparities. Our exploration draws on theoretical foundations and empirical evidence, grounding itself in the wealth of existing literature that not only informs our comprehension of political trust but also navigates the intricacies of conceptualization and measurement. Building on the foundations established in the introduction, our study delves into the rich tapestry of existing literature, theoretical frameworks, and policy debates, particularly concerning the intricate relationship between external crises, such as pandemics, and political trust. As we navigate this complex terrain, our research endeavors to contribute valuable insights by seamlessly integrating contextual, conceptual, and theoretical dimensions. Through a comprehensive analysis, we aspire to illuminate the dynamic landscape of political trust during the COVID-19 pandemic, thereby enhancing our understanding of the intricate dynamics at play in the wake of this unprecedented global challenge.

4. DATA DESCRIPTION

Firstly, this research uses the data from the dataset 'europecovid.dta' which contains a set of variables about COVID-19. The variables used in cluster analysis about the infection are 'total_deaths' and 'total_cases', two metric variables ranging from 28 to 93317 and from 5743 to 3159297 respectively, while the variables about policy responses are 'stringency_index' and 'total_tests' extending from 36.6 to 65 and from 241454 to $9.11e+07$ respectively. The variables concerned with infections have a total absolute frequency of 37 observations, those referring to policy responses have 35 observations. Eventually, to profile clusters we have used the numeric variables 'Life_expectancy', 'Extreme_poverty' and 'Median_age'. The first ranges from 72.06 to 83.78 and has 36 observations; the second ranges from 0.1 to 5.7 and has 28 observations; the third ranges from 37.3 to 47.9 and has 36 observations. In the second

part, we also use data from the ESS, an academically driven cross-national survey that has been conducted throughout Europe since 2001. Every two years, in-person interviews with carefully selected representative samples are conducted as part of this survey. To conduct our analysis, we merge data from the dataset 'europecovid.dta' and ESS round 10 which was conducted in 2020. In order to gather a representative sample of participants from European nations, the ESS uses a sophisticated multistage probability sampling technique. Trained interviewers conduct in-person interviews with members of specific households. Households and people are selected at random within each cluster that has been determined (using cluster analysis). Interviewer training, supervision, and monitoring—all of which are meticulously handled by the ESS—ensure the quality of the data in terms of validity, reliability, accuracy, and precision. The dependent variables we have selected to capture institutional trust in the ESS are: Most people can be trusted or you can't be too careful 'Ppltrst', Trust in scientists 'Trstsci', Trust in country's parliament 'Trstprl', Trust in the legal system 'Trstlgl', Trust in the police 'Trstplc', Trust in politicians 'Trstplt', Trust in political parties 'Trstprt', Trust in the European Parliament 'Trstep', Trust in the United Nations 'Trstun'. All these variables have been measured through an anchored scale with a range from 0 to 10, where the value 0 designates 'No trust at all' and value 10 stands for 'Complete trust'. Through a factor analysis we have grouped these variables into the factors: Factor 1 'Politics', Factor 2 'int_inst', and Factor 3 'local_law'. The explanatory variables we are using in the regression analysis are the five clusters of countries obtained through the cluster analysis performed.

5. MULTIVARIATE ANALYSIS

5.1. First part: CLUSTER ANALYSIS

Cluster analysis is a multivariate statistical technique that makes it possible to group cases on the basis of measured characteristics (variables). The reasons why we decided to use such approach to our analysis is because cluster analysis makes it possible to classify cases both for explanatory purposes and for confirmatory purposes. In addition, classifying large number of observations into groups allows us to understand relationships that otherwise would not be evident. In order to perform cluster analysis we have gone through 8 main steps. First of all, we select the clustering variables: *total_cases*, *total_deaths*, *total_tests* and *stringency_index*. Then we proceeded with the data screening stage, looking at missing values and since we have found two missing values for the variables *total_tests* and *stringency_index* we decided to drop them. To clean our data we also looked at outliers that are aberrant observations not representative of the general population. Since we have found one outlier - Russia, as shown in Exhibit 1, we decided to remove it. Producing again the heatmap, as shown in Exhibit 2, another outlier came up – Great Britain, and we decided to drop it as well to have cleared data. Since we have reduced the range of dissimilarity from (8.0e+15, 2.8e+14) to (1.2e+15, 4.3e+13), we decided to handle any other outliers using the appropriate linkage algorithm.

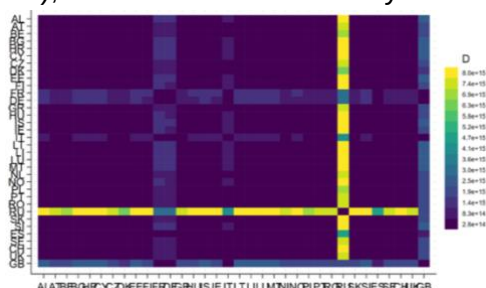


Exhibit 1: Heatplot 1

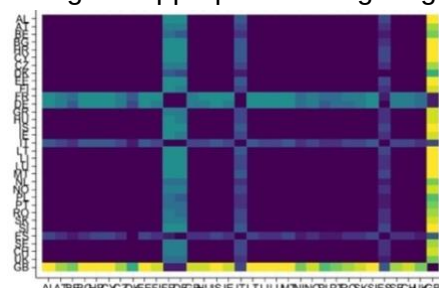


Exhibit 2: Heatplot 2

Furthermore, since cluster analysis is very sensitive to different scales and magnitudes, we looked at the values of our variables and we concluded it was appropriate to standardize them before actually performing the analysis. Once the data screening stage has been completed, we select Euclidean distance since we have standardized our variables. This approach measures the shortest distance between two points and; it is also called straight line distance. For the clustering procedure we adopted a hierarchical one from which we'll develop a dendrogram (Exhibit 3). Another choice we had to make is related to the clustering algorithm

that defines how similarity is defined between multiple member clusters in the clustering process. We went for the average linkage algorithm as with this method we are able to generate clusters with small within-cluster variation and less affected by outliers.

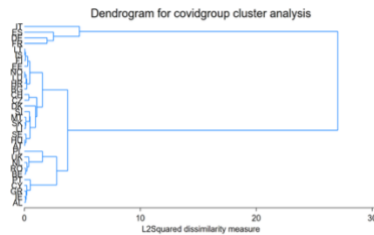


Exhibit 3: Dendrogram

Cluster No.	Country Codes				
1	AL	CY	GR	AI	PT
2	BE	NL	PL	RO	UK
3	AT	BG	CH	CZ	DK
	EE	FI	HR	HU	IS
	LI	LT	LU	MT	NO
	SE	SI	SK		
4	DE	ES	FR		
5	IT				

Exhibit 4: Clusters by Country codes

In order to select the number of clusters, we used the two stopping rules available: Calinski-Harabasz and Duda-Hart. For both of them, larger values indicate more distinct clustering. Looking at their values, we were considering whether to select 2 or 5 clusters; having performed our analysis on 2 we saw that one cluster was unproportionally big with respect to the other, as a result we selected 5 clusters. Finally, we characterized the clusters, as shown in the Exhibit 5. From the latter we can see how our clusters differ in term of the variables. Countries belonging to cluster 4 and 5, show similar characteristics. More specifically, they show high incidence for cases, deaths and tests and they have strict covid measures, especially for Italy. Countries belonging to cluster 3 show negative values for all the variables in our analysis. Cluster 1 shows similar pattern to cluster 3 except for stringency index which is instead positive. Countries belonging to cluster 2 show positive values for deaths, cases and strength of covid measures but at a much lower rate compared to countries belonging to cluster 4 and 5.

Covid_clust2	total_deaths_sd	total_cases_sd	total_test_sd	string_index_sd	median_age_sd	life_expectancy_sd	extreme_poverty_sd
1	-.51	-.58	-.42	1.02	41.10	81.23	.83
2	.27	.53	-.04	.43	42.24	78.15	2.00
3	-.47	-.49	-.42	-0.70	42.44	79.98	.54
4	2.16	2.22	2.59	.93	44.70	82.52	1.00
5	3.13	2.39	2.09	2.59	47.90	83.51	2.00
Total	-2.79e-0.9	1.86e-0.9	-4.31e-0.9	1.69e-0.9	42.58	80.24	.86

Exhibit 5: Cluster summary statistic and profiling

As last step, we went through the profiling stage in which we describe the clusters with respect to variables that have not being involved in the analysis. We have decided to profile demographic and psychographic variable; more specifically we have added life expectancy, median age and extreme poverty, as shown in Exhibit 5. Italy is the country with the highest median age, life expectancy and poverty. Countries belonging to cluster 3 are the ones showing the lowest level of poverty and a life expectancy of around 80 years. The lowest life expectancy is found in those countries belonging to cluster 2.

5.2. Second part: FACTOR ANALYSIS AND REGRESSION

The second part of our analysis involves the use of factor analysis; an interdependence model technique. In this case, within the set of variables at our disposal, we want to identify a subset of the that are strongly correlated among each other and weakly correlated with other variables. In order to capture trust in the dataset we selected the variables *ppltrst*, *trstprl*, *trstlgl*, *trstplc*, *trstplt*, *trstprt*, *trstep*, *trstun*, *trstsci*. Between the two methods available to perform factor analysis, the approach we used to summarize the data was through Principal Axis Factoring (PAF), which focuses on factoring only the common variance. The first step of our analysis was to create the correlation matrix among these variables, as shown in **Exhibit 6**. After that, we looked at the three main assumptions needed to perform the Factor analysis. We obtained a determinant equal to 0.002 and we can conclude that the first assumption is met. In order to reach reliable results, the objective is to obtain a determinant that is close to zero but different from it. This assumption is important because a determinant equal to zero would imply that the variables we are using are perfect collinear. Secondly, our analysis showed a KMO = 0.885 which is a very good result since it measures how well each variable is predicted by the other.

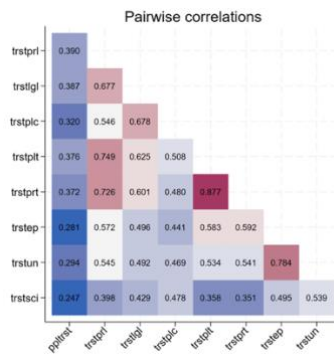


Exhibit 6

Factor	Variance	Difference	Proportion	Cumulative
Factor1	2.38	0.56	0.45	0.45
Factor2	1.83	0.24	0.34	0.79
Factor3	1.58	1.58	0.30	1.09
Factor4	0.00	.	0.00	1.09

Variable	Factor1	Factor2	Factor3	Factor4	Uniqueness
poltrst	0.30	0.16	0.31	0.03	0.78
int_inst	0.65	0.33	0.44	0.03	0.28
local_inst	0.45	0.26	0.66	-0.00	0.30
local_inst	0.28	0.31	0.65	-0.01	0.40
local_inst	0.82	0.29	0.31	-0.01	0.14
local_inst	0.82	0.31	0.27	-0.01	0.16
local_inst	0.38	0.73	0.21	0.00	0.28
local_inst	0.30	0.75	0.26	0.01	0.28
local_inst	0.12	0.50	0.40	-0.00	0.57

Exhibit 7

Finally, according to the Bartlett test of sphericity that evaluates the overall significance of the correlation matrix we can reject the null hypothesis since the p-value=0. From the correlation matrix and from the heatmap, it is possible to see that these variables have a reasonably strong association. This leads us, according to the Latent Root Criterion, to choose factors for which the eigenvalue is greater than 1, which left us with three factors. We accompanied our analysis performing an orthogonal rotation that allowed us to manage more interpretable results as shown in **Exhibit 7**. Factor 1 explains the most trust in politicians and political parties and we named it *politics*. Factor 2 explains the most trust in European parliament and UN and we named it *int_inst*. Factor 3 explains the most trust in legal system and police and we named it *local_inst*. As a final step, in order to make an assessment on whether there is significant difference in trust by those clusters we identified when performing cluster analysis, we perform three different regression models. The dependent variables are represented by the factors we created above, while the independent variables are the countries grouped into clusters. As a final step, in order to make an assessment on whether there is significant difference in trust by those clusters we identified when performing cluster analysis, we perform three different regression models. The dependent variables are represented by the factors we created above, while the independent variables are the countries grouped into clusters. **Exhibit 8** shows that there is significant difference in trust in politics and in local law by all the clusters. We cannot conclude the same thing for the factor representing trust in international institutions. Furthermore, in order to analyze better our results, we can see that, the country belonging to cluster 5, Italy, that is the country that previously showed highest level of deaths, tests, cases and stringency index, is the one that has the most negative relationship with trust. The last conclusion we find extremely interesting to report is the comparison between cluster 1 and cluster 3. In the cluster analysis above, both countries show that they have been less impacted by Covid compared to the others. However, the difference in trust in local institutions can be explained by the stringency index. The more the institutions had been stringent in terms of regulations introduced during the Covid period (e.g. lockdowns), the higher the trust in the population.

	politics	int_inst	local_inst
1. covid_cl~2	0 (.)	0 (.)	0 (.)
2. covid_cl~2	0.45*** (21.60)	0.00 (0.12)	0.07*** (3.67)
3. covid_cl~2	0.13*** (9.37)	-0.01 (-0.45)	-0.17*** (-14.54)
5. covid_cl~2	-0.13*** (-6.20)	-0.04* (-2.10)	-0.05** (-2.76)
_cons	-0.09*** (-7.42)	0.02 (1.75)	0.12*** (11.91)
N	24353	24353	24353

Exhibit 8

6. CONCLUSION

To analyze the data, we conducted a cluster analysis using variables that represent the impact of and response to COVID-19, resulting in the identification of five distinct clusters. Subsequently, we performed a factor analysis to explore the relationship between COVID-19's impact and levels of trust, focusing specifically on factors related to trust. This analysis revealed three distinct factors: trust in political systems, international institutions, and local legal and police systems. To determine whether trust levels varied based on cluster membership, we executed a regression analysis. This confirmed our hypotheses: a higher impact of COVID-19 correlates with lower trust levels in various countries, and more significant government intervention during the crisis is associated with increased trust values.

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